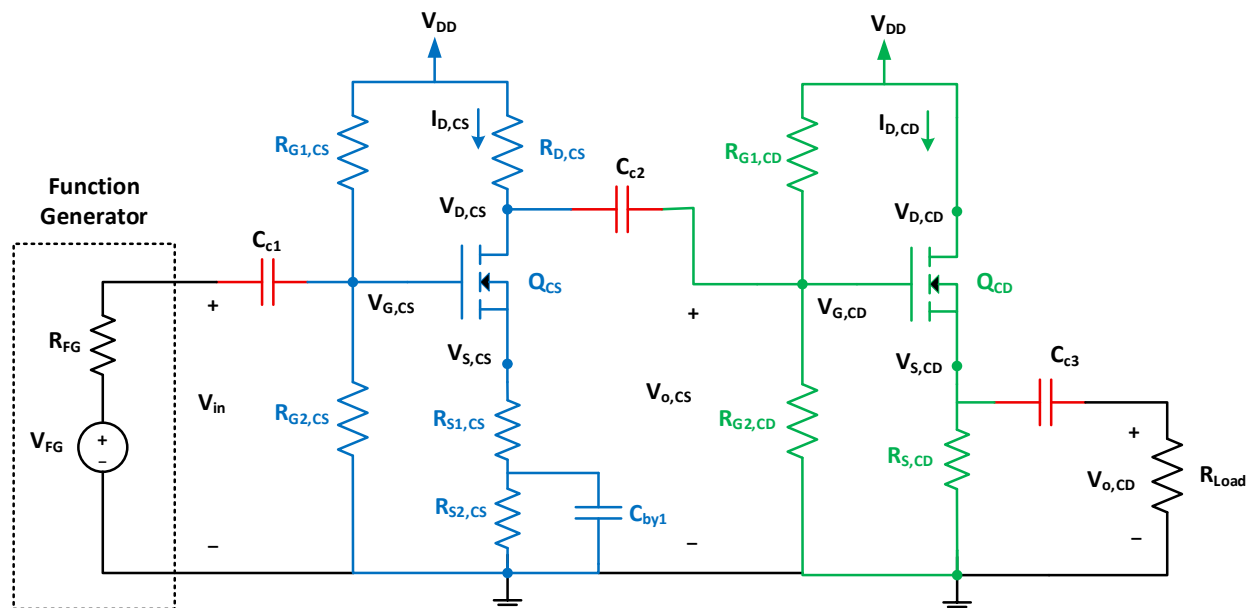


ECE321 Lab 3: Multi-Stage FET Amplifier

Goal: In this lab, you will gain experience designing a multi-stage amplifier to meet specifications and satisfy constraints. You will perform design using equations derived from small-signal and large-signal MOSFET models, verify performance using Multisim simulation, build and instrument the amplifiers in the laboratory, and collect data to verify that your design meets the specifications while performing any required troubleshooting and circuit modification.

Architecture: Your final product should track the circuit below which includes a Common-Source (CS) amplifier (blue) followed by a Common Drain (CD) amplifier (green). Coupling capacitors (red) link in the source, the stages, and the load.



Specifications & Constraints: It is understood that all parts and measurement equipment should originate from the 2G-24 laboratory. In addition, it is specified that:

1. The DC power supply voltage is constrained to be: $10V \leq V_{DD} \leq 20V$. The same voltage must be supplied to each stage.
2. The coupling (C_{c1} , C_{c2} , C_{c3}) and bypass capacitors (C_{by1}) must be electrolytic and be in the range $10\mu F \leq C \leq 100\mu F$ with suitable rated voltages.
3. All resistors should be rated 1/4W unless strongly justified otherwise.
4. The transistor must be the ZVN2106 NMOS. Absent curve tracer data, assume typical parameter values: $K_n = 80\text{mA/V}^2$, $V_t = 2.0\text{V}$, and $V_A = 100\text{V}$.

5. The amplifier must be designed, built, and verified in stages, then integrated and tested.
6. The Function Generator (FG) will serve as the input and will be set to 1kHz, Hi Z output, with the peak-to-peak amplitude adjusted as required. It can be assumed that the output resistance of the FG is 50Ω .
7. **No potentiometers** should be used to realize any of the 8 required resistors. Further, each resistor must be realized by using no more than 2 standard resistors, putting them either in series or parallel.

The remaining specifications/constraints apply to each stage:

CS-Amplifier Stage

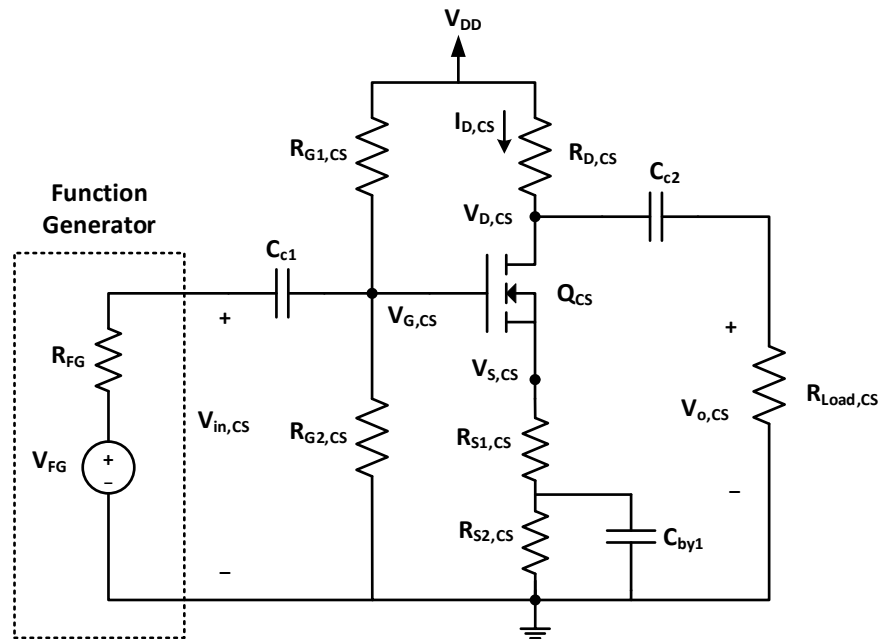
8. The gain with a $100k\Omega$ load must satisfy: $-15 V/V \leq A_v \leq -30 V/V$. The expected load resistance for this stage must satisfy: $R_L \geq 100k\Omega$.
9. The drain current must satisfy: $I_{D,CS} \leq 5mA$.
10. The input resistance must satisfy: $100k\Omega \leq R_{i,CS} \leq 150k\Omega$.
11. The output resistance must satisfy: $R_{o,CS} \leq 5.1k\Omega$.
12. The input signal must be capacitor-coupled into the stage.
13. The output voltage must be capable of achieving at least $3.2V_{pp}$ with no visible distortion.
14. The power dissipation for each part must satisfy: $P_D \leq 0.75 \times P_{rated}$.

CD-Amplifier Stage

15. The load resistance will be $R_{Load} = 200\Omega$.
16. The output voltage should reach a maximum voltage of $3V_{pp}$ with no visible distortion (obvious clipping, flattening, or pinching).
17. The gain with a 200Ω load can be anticipated to be around: $A_v \leq 0.85 V/V$.
18. The multistage gain should satisfy $-12 V/V \leq A_v \leq -30 V/V$.
19. The input resistance must satisfy: $R_{i,CD} \geq 100k\Omega$.
20. The output resistance must satisfy: $R_{o,CD} \leq 100\Omega$.
21. The drain current must satisfy: $I_{D,CD} \leq 50mA$.
22. The drain-to-source voltage must satisfy: $V_{DS,CD} \leq 11V$.
23. The power dissipation for each part must satisfy: $P_D \leq 0.75 \times P_{rated}$.
24. The Total Harmonic Voltage of the output voltage (THD_v) must be $< 5\%$ with bonus points awarded for less than 3% (and for the lowest THD_v for the section).

Day 1 Expectations:

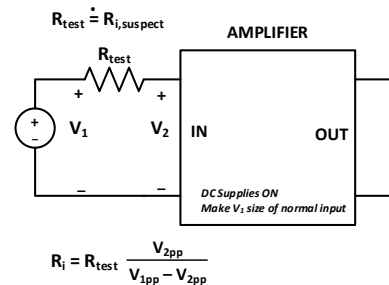
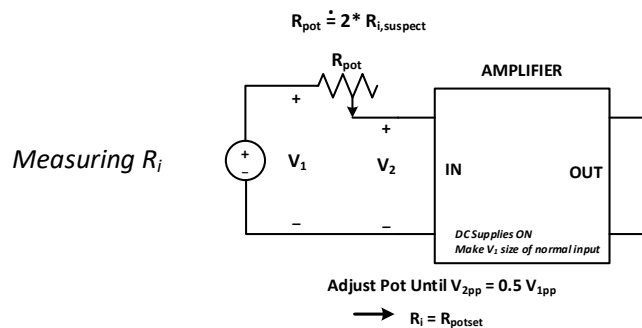
- Arrive with a CS-amplifier design that has been built, tested, and verified *in Multisim* with a load resistance of 100k Ω . Have the design documented. Know the required polarities for C_{c1} , C_{c2} , C_{by1} .



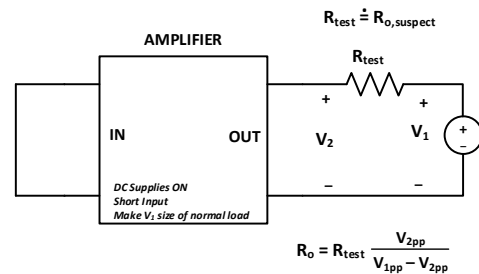
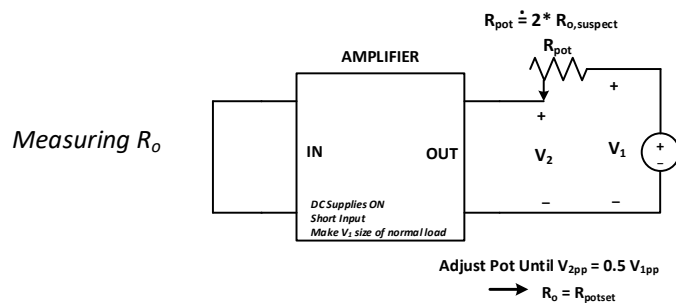
- Build the circuit in the lab adhering to the aforesaid requirements on the power supply, function generator, capacitors, resistors, and transistor
 - Measure and record all resistor values
 - Being mindful that you will later couple in a second stage, layout your circuit such that it is easy to instrument and troubleshoot. Make efficient use of power supply and ground rails, and avoid using unnecessarily long wires, the same color for all wires, and convoluted wiring paths
- Verify the DC operating point: $I_{D,CS}$, $V_{D,CS}$, $V_{S,CS}$, and $V_{G,CS}$ with the DMM. Note, the drain current can be found by measuring the resistance across $R_{D,CS}$ [at altitude, with a very dry environment, we naturally build up static electricity that can inadvertently damage a transistor. The gate voltage being too low is a common indicator that the gate insulation has been damaged.]

*** **Demonstrate your output for the instructor:** _____

- Show that the DC operating point changes if C_{c1} or C_{c2} are replaced by a wire.
- Verify the gain with the oscilloscope; measure it separately for output voltages of about $1.5V_{pp}$, $2.5V_{pp}$, and $3.5V_{pp}$.
- If your DC operating point and gain results are significantly different from your Multisim simulation, consider tweaking your simulation parameter values V_t and K_n until you see better correspondence.
- Investigate and record how the gain changes as the load resistance is changed between $100k\Omega$, $10k\Omega$, $1k\Omega$, and 200Ω . Keep the input voltage around $100mV$. Explain using the small-signal model why this variation is happening.
- Verify the input resistance spec using one of the two methods described below



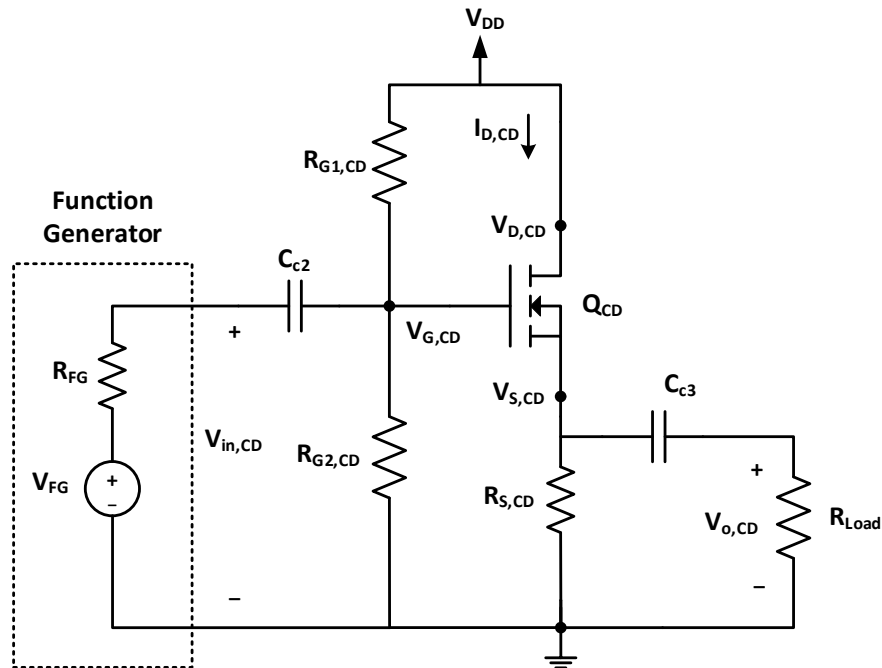
- Verify the output resistance spec using one of the two methods described below



*** **Demonstrate your output for the instructor:** _____

Day 2 Expectations:

- Arrive with a CD-amplifier design that has been built, tested, and verified *in Multisim* with a load resistance of 200Ω and an input voltage of sufficient amplitude to give a $3V_{pp}$ output voltage. Know the correct polarities for C_{c2} and C_{c3} .



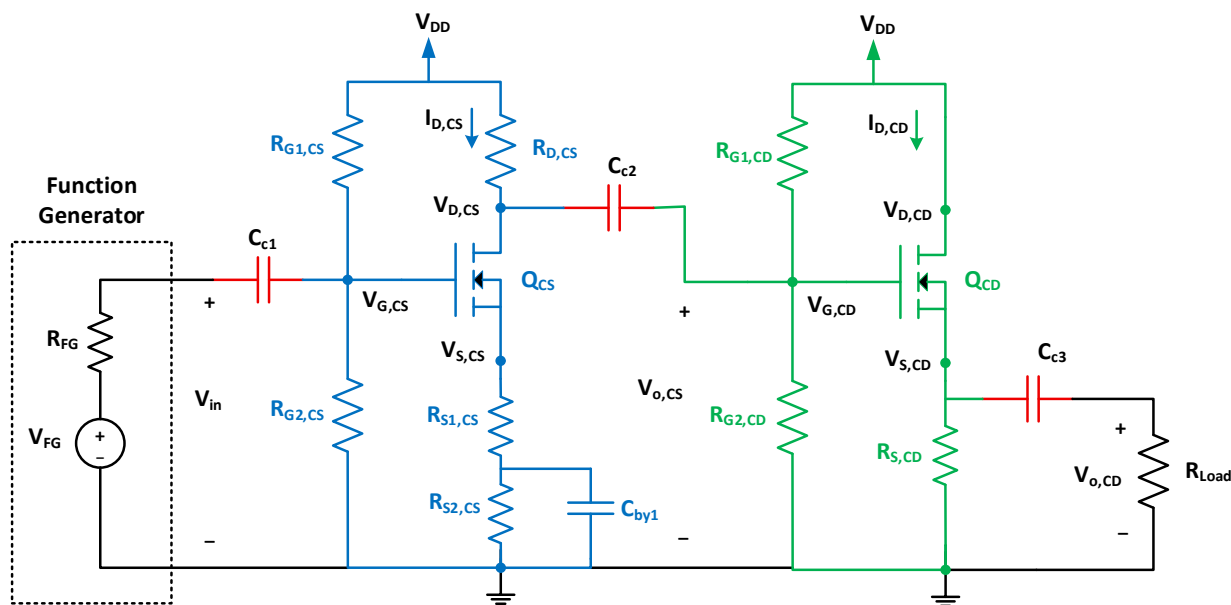
- Build the circuit in the lab adhering to the same recommendations as before. **DO NOT** couple it to the CS-amplifier. Use the FG as the input.
- Verify the DC operating point $I_{D,CD}$, $V_{D,CD}$, $V_{S,CD}$, and $V_{G,CD}$ with the DMM.

***** Demonstrate your output for the instructor:** _____

- Measure the gain with the oscilloscope; measure it for an output voltage of $3.0V_{pp}$ (*adjust the FG input amplitude accordingly*).
- Verify the output resistance spec (as described previously). Note, this measurement is *historically difficult* because our output resistance should be small meaning the FG will be loaded.
- Note, we will not measure the input resistance as our CS-amplifier design and measurement should give us sufficient confidence here.

Day 3 Expectations:

- Arrive with an integrated CS-amplifier + CD-amplifier design that has been built, tested, and verified *in Multisim* with a load resistance of 200Ω and an input voltage of sufficient amplitude to give a $3V_{pp}$ output voltage.
- You should now know the DC bias voltages $V_{D,CS}$ and $V_{G,CD}$ and should be able to identify the correct polarity for C_{c2} .



- Couple the CS-amplifier to the CD-amplifier via capacitor C_{c2} .
- Confirm that the DC bias points remain as expected using the DMM (I_D and V_{DS} are sufficient for each stage)
- Adjust the FG to achieve an output voltage across the 200Ω load of $3V_{pp}$. Make sure that the output voltage has no visible distortion, then measure the gain and verify that it meets the specification. **Print off the scope image showing input and output voltage and their measurements.**

*** **Demonstrate your output for the instructor:** _____

- Measure and record your output voltage THD using the distortion analyzer (instructions in this appendix). Verify that it meets specifications. Take a clear readable image of the analyzer output.

***** Demonstrate your output for the instructor:** _____

- Time-permitting, consider a redesign of the CS-amplifier stage to lower your achieved THD_v level. Document your results.

Follow-Up Questions:

1. Using Multisim only, describe and investigate how you can accommodate a smaller value of R_{Load} . How small can you go?
2. Use Multisim to verify the power dissipation for each component (you can ignore the resistors connected to the gate because if you selected them properly, these power levels should be quite small). Note, the power dissipation for the transistor is found from $P_D = V_{DS} \times I_D$. Discuss what to do with the AC variation.
3. Draw your multistage amplifier using the basic amplifier model (R_i , R_o , A_{vo})
4. Discuss errors between your theoretical calculations, simulations, and hardware measurements. Consider what might be the dominant sources of error. Consider things like model errors, measurement errors, and component variation.

Documentation

- All test setups and data collection should be reported in your lab notebook
- The lab report should document and summarize the investigation of each day's work. It should include answers to all questions posed. It should include simulation diagrams and any associated DC Operating Point, Transient, and THD studies that you conducted to verify your design. It should also include your design steps and a final diagram with all of the measured component values (not capacitors). Finally, you should include a discussion of whether all specifications were met and lessons learned.

Appendix: Measurement of THD Using the HP 3561A Dynamic Signal Analyzer

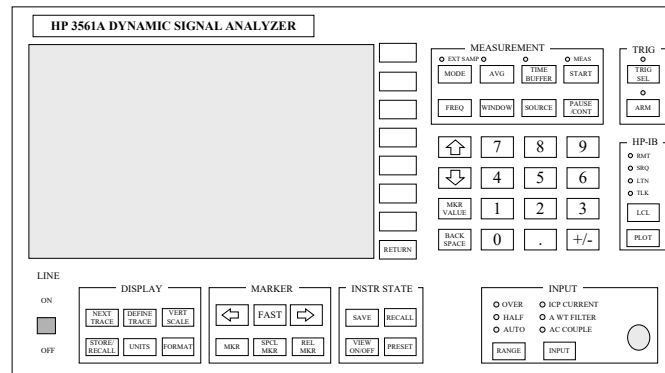


Figure 1. HP 3561A Dynamic Signal Analyzer

1. The HP 3561A Dynamic Signal Analyzer will be used to measure the Total Harmonic Distortion (THD) of your amplifier.

NOTE: You must use a single (possibly distorted) sine wave as the input signal to the signal analyzer because it will reference an internal sine wave in the analyzer itself.

2. Press "ON."
3. Apply the signal to be analyzed (V_{out} = Red Lead) into the Signal Analyzer. Make sure you connect the Black lead to GROUND = 0V.
4. Press "SPCL MKR" (special marker)
5. Press the *menu* button corresponding to "HARMONIC." The Signal Analyzer should display THD in dB at the bottom of the screen; however, the remaining steps are required for a proper THD reading in %.
6. Press the menu button corresponding to "DEFINE FUND FRQ." Use the keypad to enter the input fundamental frequency (e.g, $f_0 = 1000$ Hz for $v_{in}(t) = 2 \sin(2\pi 1000t)$ V). Press the menu button corresponding the proper units (i.e., Hz, kHz, MHz). The purpose of this step is to match your signal with the reference signal of the Signal Analyzer.
7. Press "FREQ."
8. Press the menu button corresponding to "DEFINE SPAN" and enter 10 kHz. The purpose of this step is view the fundamental and first nine harmonics in frequency domain. (Assuming you are using a 1kHz signal.)
9. Press "VERT SCALE."
10. Press the menu button corresponding to "LINEAR LOG" to obtain the THD as a percentage (rather than dB). Record the THD